Agenda

* Model Interpretability
* Feature Importance
* Gradient Descent Revision
* Calculate the Gradient $\nabla L = \begin{bmatrix} \frac{\partial L}{\partial w_1} \\ \vdots \\ \frac{\partial L}{\partial w_{2d}} \end{bmatrix}$
* Implementation

Assignments

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Recap

* Linear Reg
  ↓
  Sklearn
* Univariate Lin Reg [1 var]
  Multivariate " [>1 "]
* Evaluation metrics

✓✓ MSE = $\frac{1}{n}\sum_{i=1}^{n}[y^{(i)} - \hat{y}^{(i)}]^2$

✓✓ R2-score = $1 - \frac{SS_{Res}}{SS_{total}}$

$1 - \frac{\text{MSE of}}{\text{MSE of mean model}}$

1 → ↓ 0

{ Aprati
  1x1 ✓✓ } =>



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Model Interpretability

$$\hat{y} = w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + \cdots + w_d x_d + w_0$$

$$\Rightarrow \hat{y} \quad \cdots + (-10000)\,age + (-10)(odometer) + (200)(Engine) + \cdots$$

Case 1: $w_j \rightarrow$ -ve weight of feature

age ↑

$\hat{y}$ ↓ decrease

age → age + 1

$\hat{y}$ → $\hat{y} - 10000$

4 → 5

case 2 $w_j^\circ \rightarrow$ +ve weight of feature

engine ↑

$\hat{y}$ ↑ increase

$$\begin{cases} \text{engine} \rightarrow \text{engine} + 1 \\ \hat{y} \rightarrow \hat{y} + 200 \end{cases}$$

case 3: $w_j^\circ \rightarrow 0$

No impact on o/p variable ($\hat{y}$) due to f_j°

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Feature importance

"weights"

⇒ magnitude of weight is large, the importance of feature would be higher.

① $\hat{y} = 0.8\,x_2 + \text{-----} + 0.42\,x_5$

x_2 is higher importance

② $\hat{y} = (-1.9)\,x_2 + \text{-----} + (-4.4)\,x_5$

1.9

x_5 ✓

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③ $\hat{y} = (2.3)x_2 + \cdots + (-5.1)x_5$

x_5 >>>>>

Feature Scaling

Min Max Scaler (or) Normalization

age & odometer

[1 - 15]

[10K - 150K]

$\hat{y} = \cdots\cdots (-10000)\,age + (-10)\,odometer - \cdots\cdots$

age >>>>> odometer

Quiz #1

Feature importance in linear regression is determined by :

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A	The magnitude of the regression coefficients.	100%
B	The number of observations in the dataset.	0%
C	The correlation between the independent variables.	0%
D	The average squared difference between the predicted and actual values.	0%

End Quiz Now

Quiz #2

Consider the following Linear Regression model equation: y = 5.2x1 - 3.8x2 + 2.1x3 + 0.01x4 - 1.5 if we were to drop one feature, which one would be the best choice ?

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A	x1	0%
B	x2	0%
C	x3	0%

d. x_4 ✓

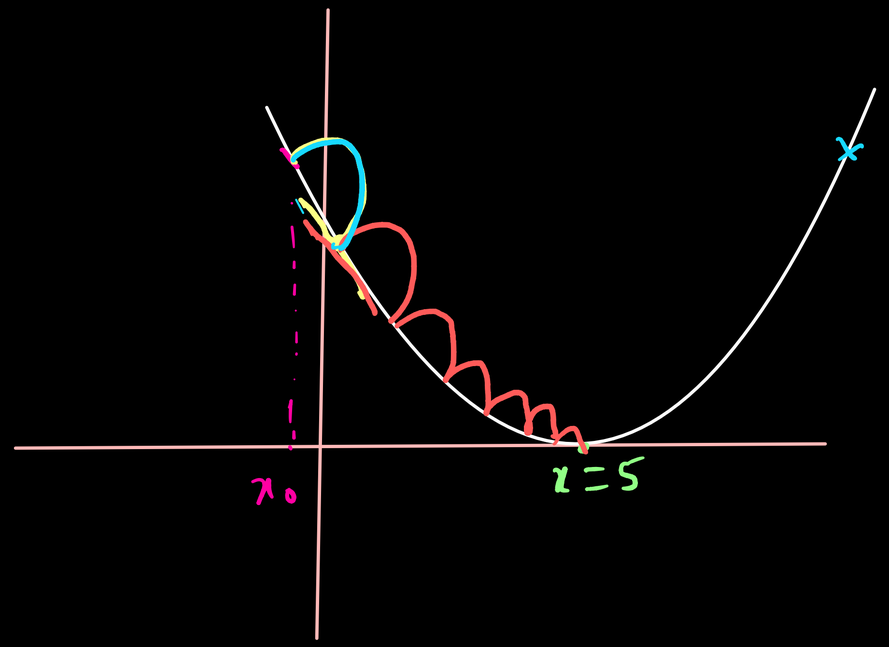
$$y = 5.2x_1 - 3.8x_2 + 2.1x_3 + 0.01x_4 - 1.5$$

Gradient Descent Algorithm

$$y = f(x) = (x-5)^2$$

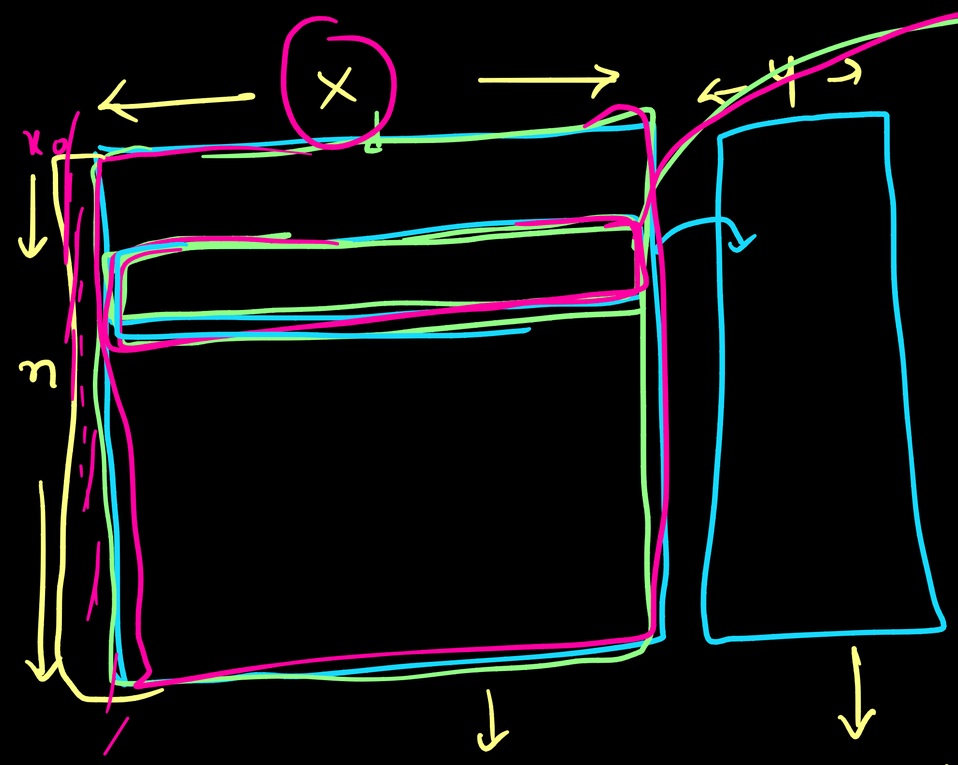
1. Pick randomly x_0
2. $\left|\frac{\partial L}{\partial x_0}\right| \rightarrow$ -ve $\rightarrow$ right
3. $x = x - \eta \frac{\partial L}{\partial x}$

η: learning rate



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Lin Reg



$$\hat{y} = w^T x + w_0$$

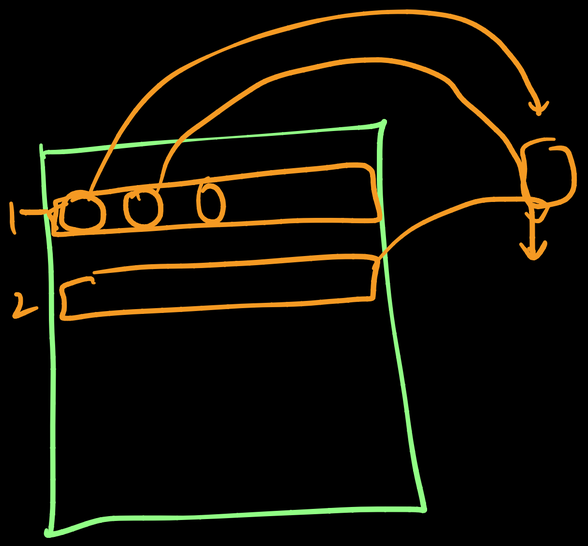
$w_1 x_1 + w_2 x_2 + \cdots w_d x_d + 1$

total no of coeff / weight $\Rightarrow$ d

$$w = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \vdots \\ w_d \end{bmatrix}$$

$(d, 1)$

$(n, 1)$



$$\hat{y} = w^T \cdot x + w_0$$

$(1, d)$ (n, d)

$$\hat{y} = x \cdot w + w_0$$

(n, d) $(d, 1)$ $(n, 1)$

$$\hat{y} = \begin{bmatrix} \hat{y}^{(1)} \\ \hat{y}^{(2)} \\ \vdots \\ \hat{y}^{(n)} \end{bmatrix}$$

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Loss function

MSE : $L = \frac{1}{n} \sum_{i=1}^{n} [y^{(i)} - \hat{y}^{(i)}]^2$

↳ G.D

optimization

$$\text{Loss}(w) = \frac{1}{n} \sum_{i=1}^{n} [y^{(i)} - \hat{y}^{(i)}]^2$$

$\hat{y}$

↳ $w_1 x_1 + w_2 x_2 + \cdots + w_d x_d$

optimization problem ⟹ ?

$\arg\min_{w}$

$$\frac{1}{n}\sum_{i=1}^{n}\left[y^{(i)} - \hat{y}^{(i)}\right]^2 \rightarrow$$

w_0

$w_1, w_2, \ldots, w_d$

1. Randomly init all 'w' vector

2. $\frac{\partial L}{\partial w_j}$

3. Repeat

{ (calculate the gradient)

$$w_j = w_j - \eta \frac{\partial L}{\partial w_j}$$

} ↗ 100

Calculate the gradient

$$\hat{y} = w_1 x_1 + w_2 x_2 + \cdots + w_d x_d$$

$$L = \frac{1}{n} \sum_{i=1}^{n} \left[y^{(i)} - \hat{y}^{(i)} \right]^2$$

$\hat{y}^{(i)}$ away

$$\frac{\partial L}{\partial w_j}$$

single data point

$$\frac{\partial}{\partial w_j} (y - \hat{y})^2$$

Chain rule

$$= \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial w_j}$$

$$= \frac{\partial}{\partial \hat{y}} (y - \hat{y})^2$$

$$\frac{d}{dx} x^2 \quad 2x$$

$$\frac{\partial}{\partial x} (1 - x)^2$$

$$2(1-x) \frac{\partial (-x)}{\partial x}$$

$$2(1-x)(-1)$$

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$$= 2(y-\hat{y})(-1) * \frac{\partial}{\partial w_j}\left(w_1 x_1 + w_2 x_2 + \cdots + w_j x_j + \cdots + w_d x_d + w_0\right)$$

$$= 2(y-\hat{y})(-1) * x_j$$

$$\frac{\partial L}{\partial w_j} = 2(\hat{y}-y) * x_j$$

$$\frac{\partial L}{\partial w_0} = 2(\hat{y}-y) * 1$$

For all data points

$$\frac{\partial L}{\partial w_j} = \frac{1}{n}\sum_{i=1}^{n} 2\left(\hat{y}^{(i)} - y^{(i)}\right) * x_j^{(i)}$$

$$= \frac{2}{n}\sum_{i=1}^{n}\left(\hat{y}^{(i)} - y^{(i)}\right) * x_j^{(i)}$$

=>

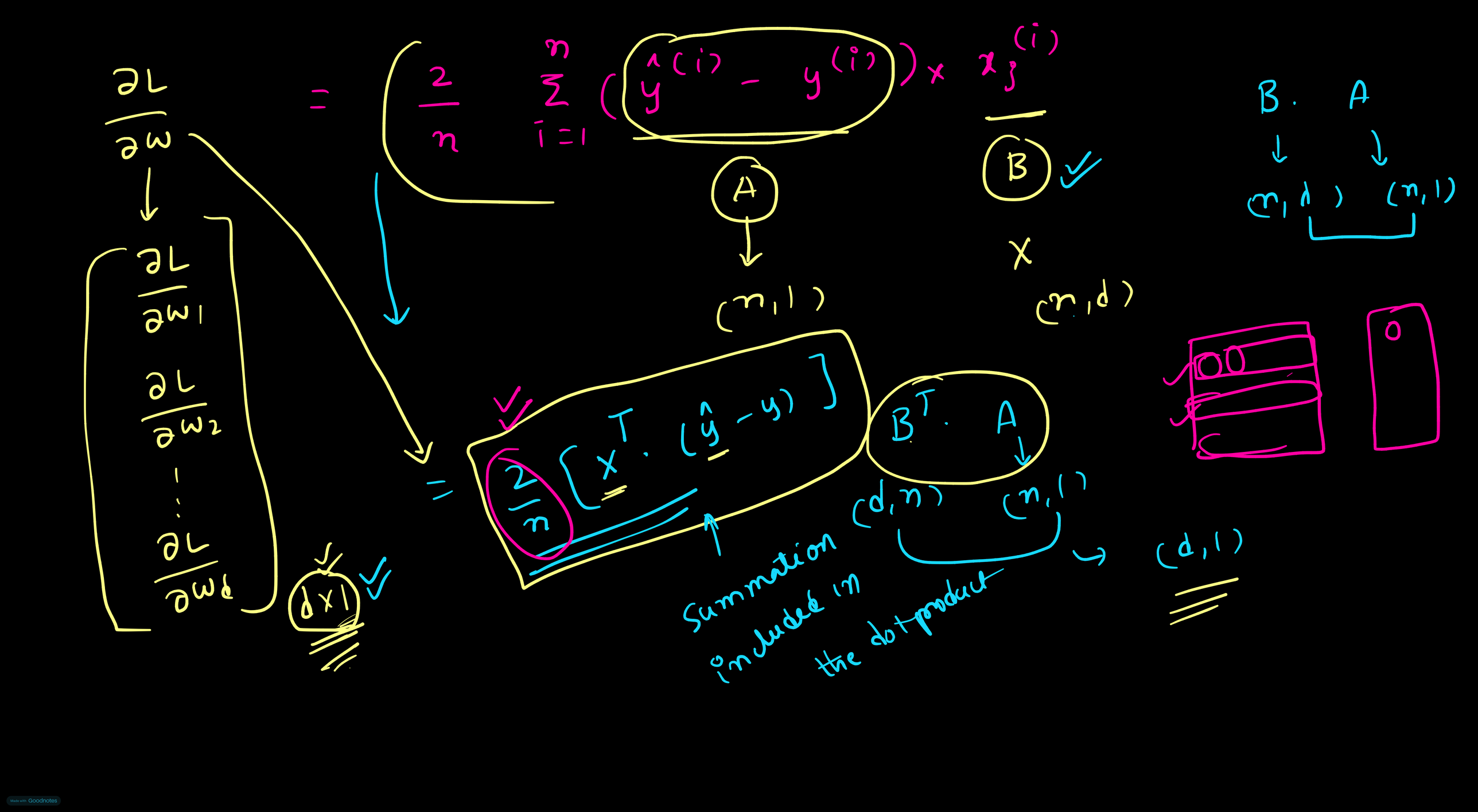
(1) Randomly init the w

(2) $\frac{\partial L}{\partial w_j}$

(3) $w_j = w_j - \eta \left(\frac{\partial L}{\partial w_j} \right)$

$= w_j - \eta \left(\frac{2}{n} \sum_{i=1}^{n} (\hat{y}^{(i)} - y^{(i)}) * x_j^{(i)} \right)$

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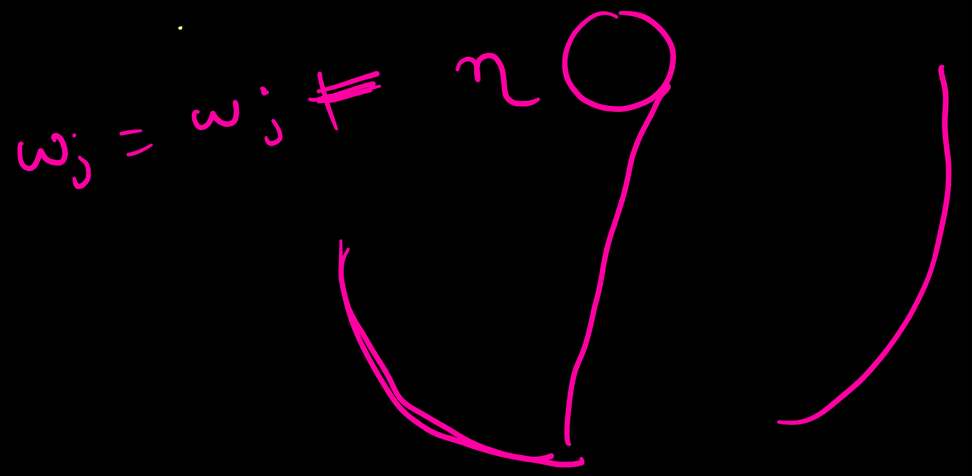

∂L/∂w = (2/n) Σ_{i=1}^{n} (ŷ^(i) − y^(i)) × x_j^(i)
A
B
X
(n, 1)
(n, d)
B . A
(n, d) (n, 1)
∂L/∂w1
∂L/∂w2
∂L/∂wd
d x 1
= (2/n) [X^T . (ŷ − y)]
B^T . A
(d, n) (n, 1)
(d, 1)
Summation included in the dot product
0

What is the objective of Gradient Descent in linear regression?

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A	Minimize the absolute error	17%
B ✓	Minimize the squared error	83%
C	Maximize the R-squared score	0%
D	Maximize the accuracy	0%

End Quiz Now



What will happen if we add the value instead of subtracting from the original weight?

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A ✓	The model will diverge instead of converging towards the optimal solution.	0%
B	The model will convergse, but very slowly	0%
C	The model will converse very fast	0%

End Quiz Now

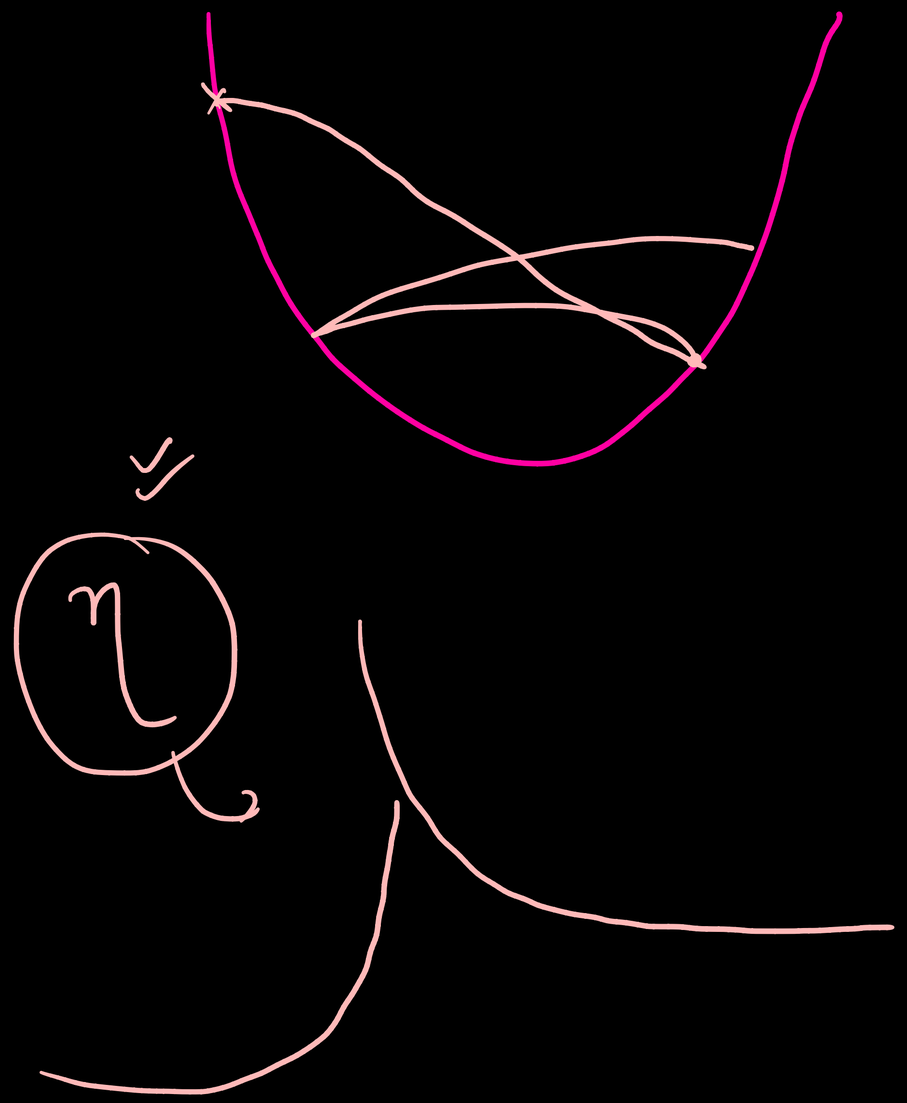
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What happens if the learning rate in gradient descent for linear regression is set too large?

0 users have participated

A	The algorithm will converge faster to the optimal solution.	0%
B	The model will overfit the training data, leading to poor generalization.	0%
C	The algorithm may fail to converge, and the coefficients may oscillate or diverge.	0%
D	The cost function will be overestimated, resulting in an inflated R2 score.	0%

End Quiz Now



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